Development of New Synthetic Methods and Strategies for the Total Syntheses of Pumiliotoxin A Class Alkaloids

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Abstract: *New strategies for the syntheses of the pumiliotoxin (PTX) A dendrobatid alkaloids have been developed which are based on palladium-catalyzed carbonylation, palladium-catalyzed cross-coupling reaction, and nickel–chromium-mediated cyclization. Application of these methodologies to the asymmetric total syntheses of (–)-PTX 209F, (–)-PTX 225F, (+)-PTX A, (+)-allo-PTX 267A, (+)-allo-PTX 339A, and (+)-homo-PTX 223G is described.*

1. Introduction

Neotropical poison-dart frogs of the Dendrobatidae family have been a rich source of a remarkable variety of alkaloids with structurally unique features and biological significance (ref. 1). During the past 30 years, more than 400 alkaloids of over 20 structural classes have been detected (ref. 1c) in skin extracts from Dendrobatidae and hence are referred to as "dendrobatid alkaloids". The pumiliotoxin A alkaloids, one of the major classes of the dendrobatid alkaloids, are a group of more than 40 alkaloids characterized by the 6-alkylidene-8-hydroxy-8-methylindolizidine ring system, which have cardiotonic and myotonic activity apparently enhancing sodium channel function (ref. 1b). They have been divided into the three subclasses, the pumiliotoxins (**1**), the allopumiliotoxins (**2**), and the

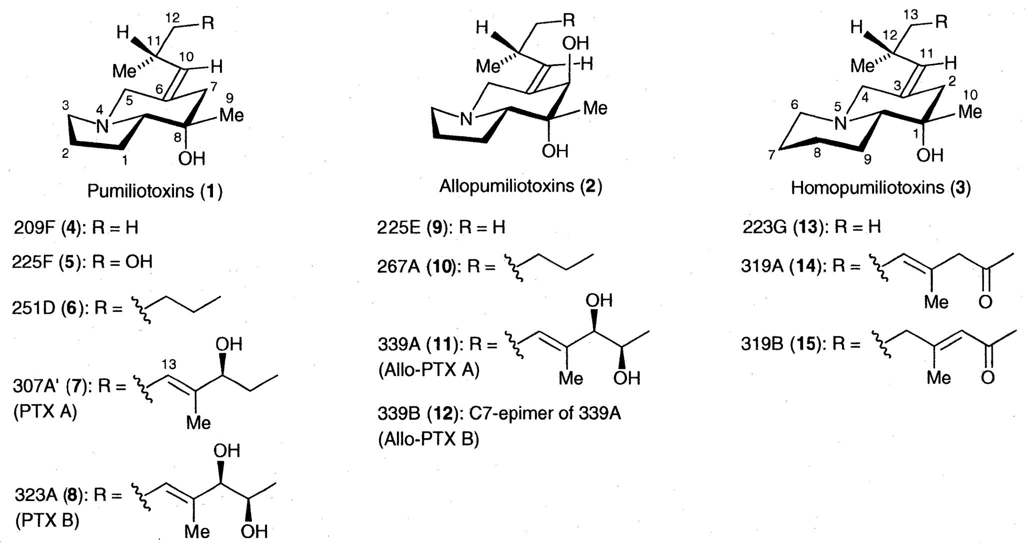


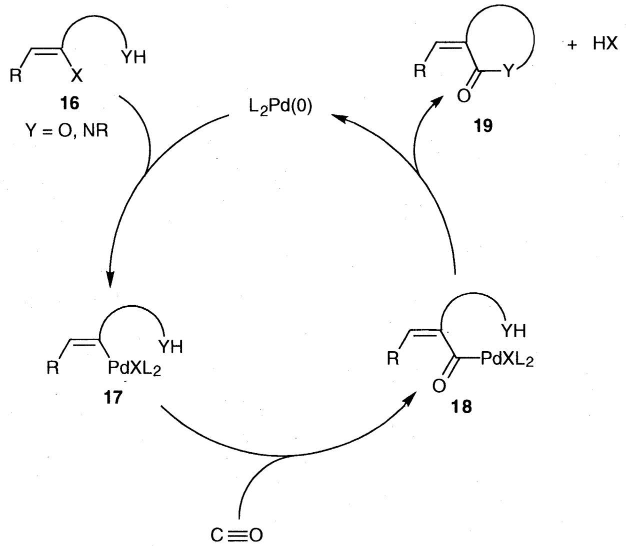
Figure 1. Representative pumiliotoxin A class alkaloids.

homopumiliotoxins (**3**) which are pumiliotoxin homologues in which the indolizidine moiety of pumiliotoxins is replaced with a quinolizidine ring (Figure 1). Around the late 1980s, an international treaty to protect endangered species was enacted (ref. 2) that has prevented the collecting of the Dendrobatid frogs whose habitat is Central and South America. In this respect, the inability of natural sources to provide sufficient material for detailed pharmacological evaluation makes the synthesis of the pumiliotoxin A alkaloids an important and urgent goal (ref. 3). For several years, research efforts in our laboratory have been directed toward the development of new synthetic methods and strategies for the total syntheses of the pumiliotoxin A family of alkaloids. The purpose of this article is to review some of our achievements in the total syntheses of the pumiliotoxin A dendrobatid alkaloids including pumiliotoxins (**1**), allopumiliotoxins (**2**), and homopumiliotoxins (**3**).

2. Syntheses of Pumiliotoxin and Homopumiliotoxin Alkaloids via Pd-Catalyzed Carbonylation

In all pumiliotoxin A dendrobatid alkaloids, a common characteristic structural feature is the presence of the (*Z*)-alkylidene side chain at C-6 of the indolizidine nucleus (or C-3 of the quinolizidine nucleus for the homopumiliotoxins). Control of the alkylidene geometry constitutes an important synthetic challenge in total synthesis of structurally complex natural products (ref. 4). The selective generation of *E*- and *Z*- exocyclic olefins has traditionally been attained via π-bond construction employing Wittig-type reaction or aldol condensation, however, these methods have been less than satisfactory. In this regard, stereodefined generation of exocyclic olefins is a fundamental requirement for achieving the synthesis of the pumiliotoxin A class of alkaloids.

2.1 Synthetic Strategy. Our strategy for the total synthesis of the pumiliotoxin A alkaloids capitalized on the palladium-catalyzed approach involving carbonylation of vinyl halides initially investigated by Heck (ref. 5), which leads to carboalkoxylation and carboamidation in the presence of alcohols and primary or secondary amines, respectively. An intramolecular version of this catalytic carbonylation can lead to a useful synthetic design and construction of heterocycles incorporating exocyclic alkenes as presented in eq 1. Clearly, the first step of this approach is oxidative addition of the (*Z*)-vinyl halide **16** to a Pd(0) species, followed by CO insertion to form an acylpalladium(II) complex **18**. Either $Pd(PPh_3)_4$ or $PdX_2(PPh_3)_2$ (X = Cl, OAc) can be used as the catalyst for these processes, the Pd(II) species being reduced readily to a Pd(0) species by carbon monoxide under the reaction conditions. Although mechanistic details are not clear, the last step involving the lactonization (Y = O) or lactamization (Y = NR) gives rise to **19** (lactone or lactam) and regenerates a Pd(0) species. This



(1)

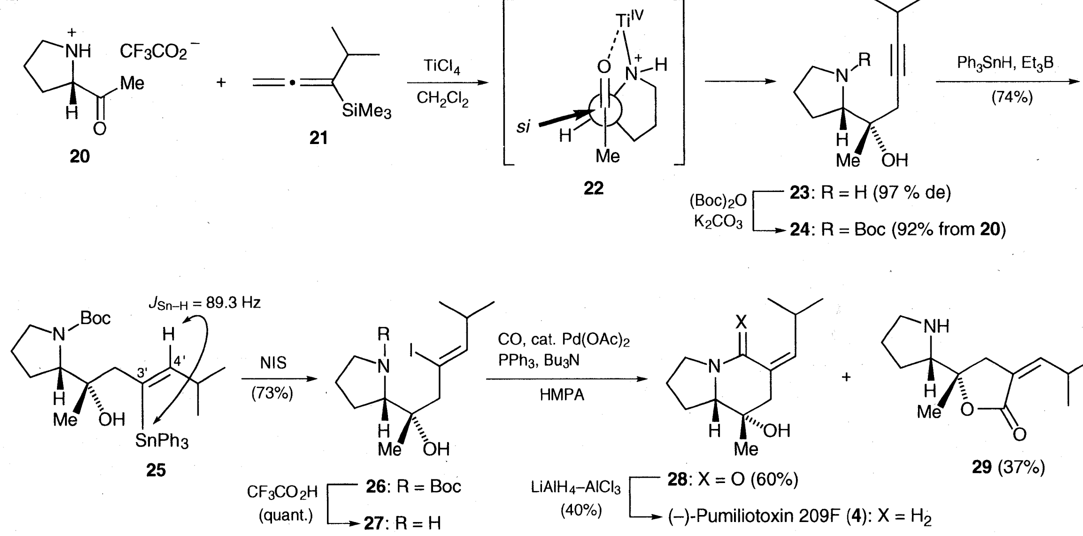
strategy takes advantage of the fact that both the oxidative addition and the CO insertion proceed with retention of (*Z*)-configuration at the vinyl carbon. We envisioned that the methodology could readily be adapted to allow the rapid construction of the pumiliotoxin framework bearing the stereodefined exocyclic alkenes.

2.2 Total Synthesis of (–)-Pumiliotoxin 209F. (–)-Pumiliotoxin 209F (**4**) is a minor dendrobatid alkaloid recently characterized as a new member of the subgroup of the pumiliotoxin alkaloids (**1**) (ref. 6). Of the 15 alkaloids so far recognized as members of this subgroup, only three alkaloids, pumiliotoxins 251D (**6**), A (**7**), and B (**8**), have been synthesized and their natural absolute stereochemistry has been established to be 8*S*,8a*S* (refs. 1, 3). No synthetic studies toward the rest of the pumiliotoxin subgroup alkaloids have been published and there is still no formal establishment of their absolute stereochemistry, though it is assumed to be the same as that of pumiliotoxins 251D, A, and B.

Our first goal was to synthesize pumiliotoxin 209F (**4**) utilizing the palladium-catalyzed lactamization of a (*Z*)-vinyl halide **27** as shown in Scheme 1 (ref. 7). This synthetic approach consists of another key step involving direct construction of a homopropargyl alcohol **23** by carbonyl addition reaction of an allenylsilane (ref. 8) as the first step of the sequence. Thus, the trifluoroacetate salt **20** of (*S*)-2-acetylpyrrolidine was treated with $TiCl_4$ and then with 1-isopropyl-1-(trimethylsilyl)allene **21** (ref. 9) in dichloromethane at –78 °C to give the homopropargyl alcohol **23** with excellent diastereoselectivity of 97% de, which was converted to its N-Boc derivative **24** in 92% overall yield from **20**. The α-facial selectivity realized in the propargylation of **20** can be rationalized by invoking a Lewis acid-chelate cyclic intermediate **22** (Cram's α-chelation model). Radical-initiated hydrostannylation of the homopropargyl alcohol **24** using triethylborane and triphenyltin hydride proceeded with complete trans selectivity to afford the (*Z*)-triphenylstannylalkene **25** along with the (*Z*)-4'-stannyl regioisomer in a 4.4:1 ratio. The assignment of the *Z* stereochemistry of **25** was made on the basis of the large coupling constant (89.3 Hz) between Sn and the vinylic proton. The regiocontrol of this hydrostannylation process would be rationalized by assuming coordination of a $Ph_3Sn^\bullet$ radical to the OH group (ref. 10) and preferential attack of the $Ph_3Sn^\bullet$ radical at the less hindered C-3'. Upon exposure of **25** to *N*-iodosuccinimide, iododestannylation proceeded with complete retention of the *Z* configuration to give the vinyl iodide **26**.

With the (*Z*)-vinyl iodide in hand, the palladium-catalyzed carbonylation was undertaken for the construction of the indolizidine framework with the (*Z*)-isobutylidene appendage. Thus, after deprotection, the amino alcohol **27** was treated with carbon monoxide and tributylamine in the presence

Scheme 1

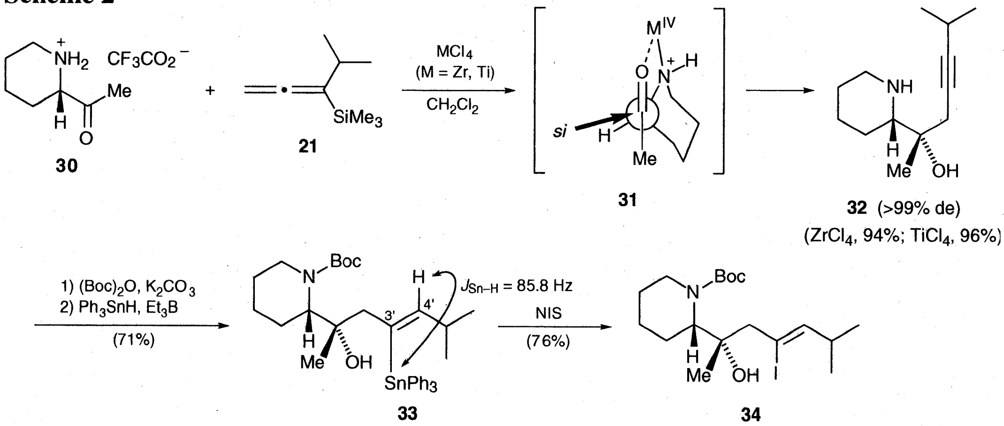


of a catalytic amount of $Pd(OAc)_2$ and triphenylphosphine in HMPA at 100 °C (ref. 11). This procedure afforded a separable mixture of the lactam **28** (60%) and lactone **29** (37%). Lactam reduction of **28** resulted in (–)-pumiliotoxin 209F (**4**) in 40% yield. Synthetic **4** displayed optical and spectroscopic characteristics identical to those reported for the natural product (ref. 6). This work constituted the first synthesis of pumiliotoxin 209F, verified the structure proposed, and established the absolute configuration of the natural product.

2.3 Total Synthesis of (+)-Homopumiliotoxin 223G. Homopumiliotoxin 223G (**13**) characterized as a quinolizidine homologue of pumiliotoxin 209F (**4**), is the first example of an unusual class of dendrobatid alkaloids with a quinolizidine ring (ref. 12). However, recent examination revealed that skin extracts from nondendrobatid amphibians, such as the new world genus of bufonid toads and the Madagascan genus of ranid frogs, also contain homopumiliotoxin 223G (ref. 1b), and the former genus contains new members of the homopumiliotoxin subclass alkaloids, e.g., 319A (**14**) and 319B (**15**) (ref. 1b). The structure with relative configuration of these homopumiliotoxin alkaloids has been determined by spectral analysis. Because of the trace quantities of the materials available for study from natural sources, their optical rotations have not determined, and their absolute stereochemistry have been only tentatively assigned as 1*S*,9a*S* based on the analogy with the established configuration of the pumiliotoxin subclass alkaloids (**1**), such as pumiliotoxins 251D (**6**), A (**7**), and B (**8**).

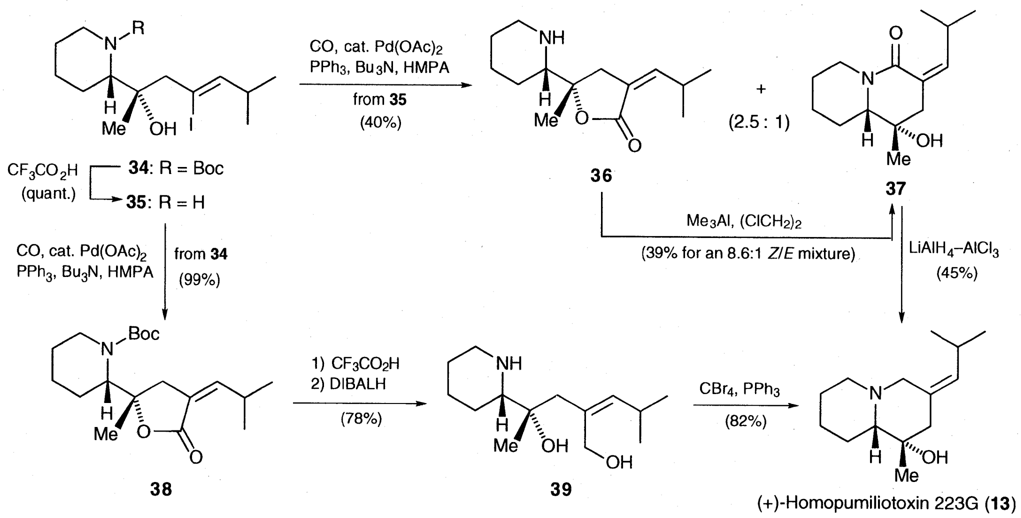
The synthesis of homopumiliotoxin 223G (**13**) (ref. 13) began with the $TiCl_4$-induced addition of the allenylsilane **21** to the trifluoroacetate salt **30** of (*S*)-2-acetylpiperidine under α-chelation control to give the homopropargyl alcohol **32** with complete stereocontrol in 96% yield (Scheme 2); no trace of the other diastereomer was detectable by NMR analysis. In this case, changing the Lewis acid from $TiCl_4$ to $ZrCl_4$ was also effective to furnish exclusively **32** in 94% yield. Subsequent radical-initiated hydrostannylation of the N-Boc derivative of **32** in a similar manner as described above for **24** led to a separable mixture of the trans adducts, the (*Z*)-triphenylstannylalkene **33** and its (*Z*)-4'-stannyl regioisomer in a 2.8:1 ratio in a combined yield of 95%. The major isomer **33** was converted via iododestannylation (NIS) to the (*Z*)-vinyl iodide **34**.

Scheme 2



After deprotection of the Boc group, the amino alcohol **35** was subjected to the palladium-catalyzed carbonylation under the same conditions as used for the amino alcohol **27** and thereby, in contrast to the cyclization of **27**, lactonization occurred in preference to lactamization, leading to a 2.5:1 mixture of the lactone **36** and the lactam **37** in a 40% combined yield. Upon treatment with trimethylaluminum, the major product lactone **36** could be transformed into the desired lactam **37** albeit as an 8.6:1 *Z*/*E* mixture and in a low yield (39%). Reduction of **37** with $LiAlH_4$–$AlCl_3$ furnished (+)-homopumiliotoxin 223G (**13**) in 45% yield. However, this sequence starting from the vinyl iodide **35** proved to be unsatisfactory, providing (+)-homopumiliotoxin 223G in a low overall yield (9%). In order to solve this problem of the preferential formation of the undesired lactone **36**, we examined an alternative approach involving selective lactonization using the N-protected vinyl iodide **34**. Upon the palladium-catalyzed carbonylation, **34** was smoothly cyclized to the lactone **38** in 99% yield, which was converted to the diol **39** via deprotection followed by DIBALH reduction. Intramolecular dehydrocyclization of **39** was performed with carbon tetrabromide and triphenylphosphine according to our previously described procedure (ref. 14) to afford (+)-homopumiliotoxin 223G (**13**), providing a remarkable improvement in the overall yield (63%) from **34**. The synthetic sample of **13** was shown, by spectral comparisons, to be identical with natural homopumiliotoxin 223G (ref. 12). The rotation observed for this product, $[\alpha]^{24}_D$ +3.45 (*c* 0.51, $CHCl_3$), suggests the rotation of natural homopumiliotoxin 223G to be dextrorotatory, although at present no data on the optical rotation of the natural product have been reported.

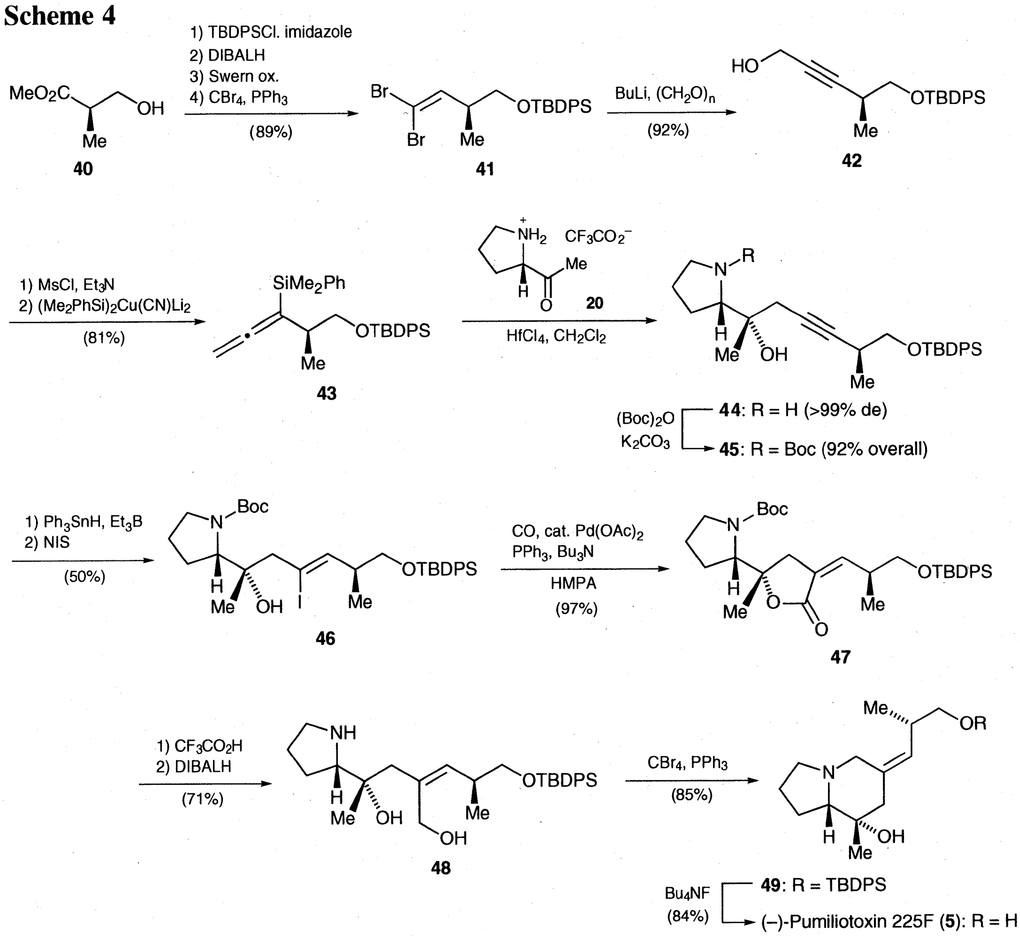
Scheme 3



2.4 Total Synthesis of (–)-Pumiliotoxin 225F. The highly efficient strategy for the construction of the (*Z*)-alkylidene bicyclic heterocyclic ring system utilizing a combination of processes involving the Lewis acid-induced chelation-controlled propargylation and palladium-catalyzed lactonization was next applied to the first synthesis of (–)-pumiliotoxin 225F (**5**) as outlined in Scheme 4 (ref. 15). For the preparation of the allenylsilane **43**, the dibromoolefin **41**, prepared from methyl (*R*)-3-hydroxyisobutyrate (**40**) in 4 steps (ref. 16), was treated with BuLi and paraformaldehyde to provide the homopropargyl alcohol **42** (92%). Construction of the allenylsilane **43** was successfully achieved using Fleming's method (ref. 17) by conversion to the mesylate and subsequent treatment with the bis(dimethylphenylsilyl)cuprate reagent (–85 °C, 3 min). Chelation-controlled carbonyl addition of the allenylsilane **43** to the trifluoroacetate salt **20** of (*S*)-2-acetylpyrrolidine proceeded cleanly by using hafnium(IV) chloride to afford the desired homopropargylic alcohol **44**, which was isolated as the N-Boc derivative **45** as a single diastereomer in excellent overall yield (92%). In this case, $HfCl_4$ was found to be the most effective Lewis acid for the nucleophilic addition (ref. 18).

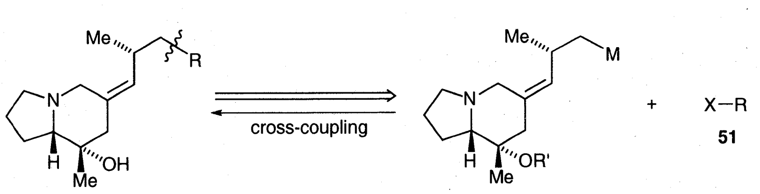
Radical hydrostannylation of **45** using triethylborane and triphenyltin hydride proceeded with complete trans selectivity to give the (*Z*)-3'-stannyl alkene (60%) along with the (*Z*)-4'-stannyl regioisomer (28%), and the former stannyl product was converted via iododestannylation followed by deprotection to afford the vinyl iodide **46** with retention of the *Z* configuration. Palladium-catalyzed carbonylation of **46** smoothly occurred to furnish the lactone **47** in 97% yield. Deprotection of **47** followed by DIBAL reduction gave the diol **48**, which underwent smooth intramolecular cyclodehydration (CBr_4, Ph_3P) to form the (*Z*)-alkylideneindolizidine **49**. Subsequent removal of the silyl protecting group resulted in the first total synthesis of (–)-pumiliotoxin 225F (**5**). The synthetic material displayed spectral properties that matched those of the natural product (ref. 6). However, the observed rotation of synthetic **5** ($[\alpha]^{24}_D$ = –25.3 (*c* = 0.25, $CHCl_3$)) was found to be significantly lower than the reported value (ref. 6) ($[\alpha]_D$ = –87.4 (*c* = 0.23, $CHCl_3$)), even though the chiral HPLC column indicated our synthetic product to be enantiomerically pure. This discrepancy may possibly be explained by the contamination with a small amount of a highly optically active impurity in the natural sample.

Scheme 4



3. Pd-Catalyzed Cross-Coupling Strategy for Pumiliotoxin Synthesis

Since the common structural motif shared by all members of the pumiliotoxin alkaloids (**1**) is the presence of the (*Z*)-alkylideneindolizidine component, the strategic disconnection of the pumiliotoxin alkaloids at the C-12–C-13 bond of the alkylidene appendage (indicated by the wavy line in eq 2) should be most appropriate for the efficient and flexible synthesis of the pumiliotoxin alkaloids. This bond



(2)

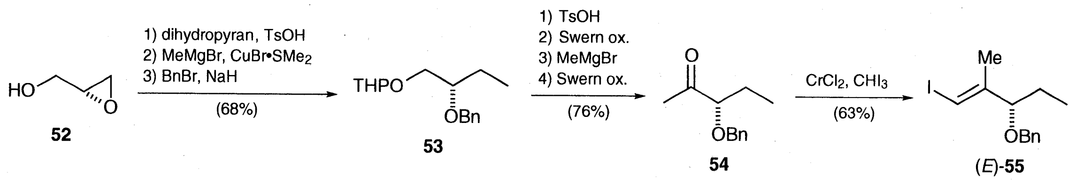
Pumiliotoxins (**1**)

50
a common intermediate for the synthesis of all members of the pumiliotoxins

would be formed by cross-coupling reaction between a metal derivative **50** of the (*Z*)-alkylideneindolizidine segment and an alkyl or vinyl halide **51**. The former segment can be utilized as a common pivotal precursor for the synthesis of the pumiliotoxins (**1**) and was expected to be readily attainable from the above-described compound **49**, a synthetic precursor of pumiliotoxin 225F (**5**). We felt that this convergent route would yield a conceptually simple and potentially more general approach to the alkaloids of the pumiliotoxin family (**1**).

3.1 Total Synthesis of (+)-Pumiliotoxin A. The side chain fragment produced by disconnection at C-12–C-13 of (+)-pumiliotoxin A (**7**) was prepared starting from (*R*)-glycidol (**52**) in 33% overall yield (Scheme 5). The key step in the formation of the vinyl iodide (*E*)-**55** (separated from an 89:11 *E*/*Z* mixture), necessary for coupling with the organometallic compound **50**, was $CrCl_2$-mediated stereoselective construction of alkenyl halides with one-carbon homologation (ref. 19).

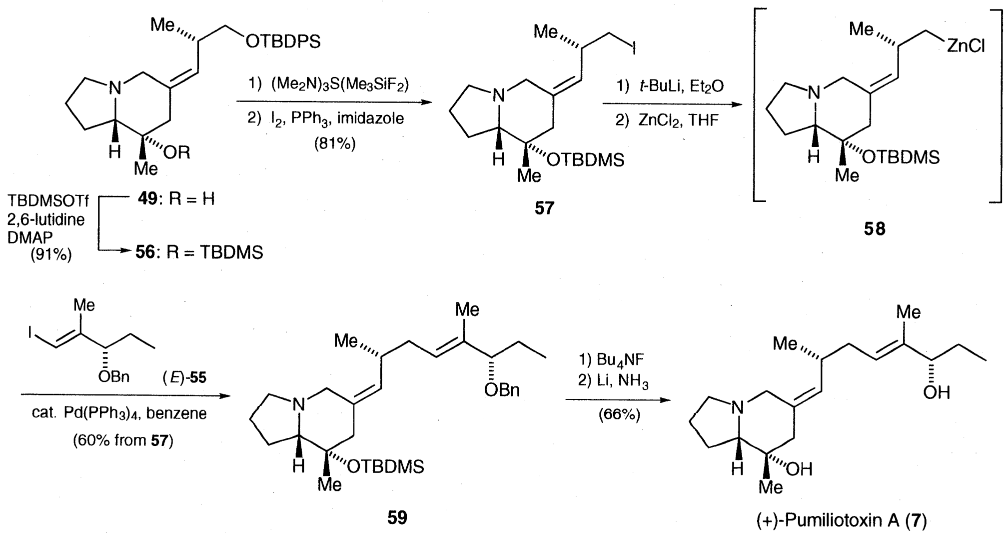
Scheme 5



The (*Z*)-alkylideneindolizidine metal species (**50** in eq 2) was next prepared from **49**, which was silylated to give **56** with the tertiary and primary alcohols protected with the TBDMS and TBDPS groups, respectively. The selective deprotection of the primary TBDPS ether was effected using tris(dimethylamino)sulfonium difluorotrimethylsilicate (TAS-F) (1.5 equiv) according to Roush's procedure (ref. 20) and provided the primary alcohol, which was then converted to the iodide **57** (I_2, PPh_3, imidazole).

The stage was then set for the critical cross-coupling reaction with the chiral vinyl iodide (*E*)-**55** for the synthesis of pumiliotoxin A (**7**). One potential problem associated with the transition metal-catalyzed homoallyl–alkenyl coupling would be the tendency of the homoallylic compounds to undergo β-elimination. This problem was overcome by Negishi (ref. 21) who adapted homoallylic organozincs to palladium-catalyzed cross-coupling with alkenyl halides to effect the construction of 1,5-dienes. In view of these results, we explored the use of the organozinc for the cross-coupling reaction (ref. 22).

Scheme 6



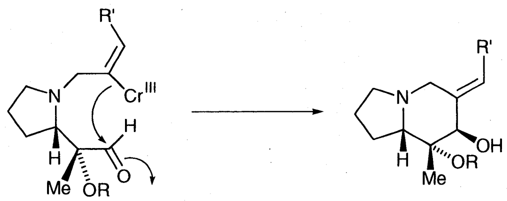
Due to the difficulty usually associated with preparation of homoallylzinc species by direct zinc insertion to the corresponding halides, the homoallyl iodide **57** was subjected to halogen–metal exchange with *t*-BuLi at –110 °C, followed by transmetalation with $ZnCl_2$. Subsequent one-pot treatment of the resulting homoallylzinc reagent **58** with (*E*)-**55** in the presence of 13 mol % of $Pd(PPh_3)_4$ in benzene at room temperature afforded the cross-coupled product **59** in 60% yield from **57** with complete retention of configuration of the stereocenter(s) and (*Z*)-geometry. Finally, removal of the TBDMS and benzyl protecting groups provided (+)-pumiliotoxin A (**7**) (ref. 15).

The new strategy developed herein has served to demonstrate the potential of the homoallyl–alkenyl coupling protocol for a general entry to the convergent asymmetric synthesis of the pumiliotoxin alkaloids. It relies on a palladium(0)-based cross-coupling reaction employing a novel, complex organozinc molecule with a nitrogen heterocycle, which is, to the best of our knowledge, the first example of the application of such reaction in natural product synthesis (ref. 23). This convergent approach utilizing the highly functionalized organozinc (**58**) as a common synthetic intermediate should be applicable to total syntheses of all members of the family of the pumiliotoxin alkaloids.

4. Synthesis of Allopumiliotoxin Alkaloids via Ni(II)/Cr(II)-Mediated Cyclization

The allopumiliotoxin class dendrobatid alkaloids (**2**) are hydroxy congeners of the pumiliotoxin class (**1**) and possess a vicinal diol group in the indolizidine ring. They are the most complex members of the pumiliotoxin A alkaloid group. In studies aimed at developing another approach to the alkaloids of the pumiliotoxin A family, we have achieved the total synthesis of allopumiliotoxins 267A (**10**) and 339A (**11**) (ref. 24).

4.1 Synthetic Strategy. As revealed by the structures, the synthesis of these alkaloids **10** and **11** poses two fundamental problems: (1) the introduction of the axially oriented vicinal dihydroxy groups at C-7 and C-8 to the indolizidine ring and (2) the construction of an exocyclic (*E*)-alkene. To overcome these problems, we envisioned an intramolecular approach based on alkenyl metal cyclization involving a chromium-mediated coupling reaction via a vinylchromium intermediate **60** (eq 3). Such reaction involving carbon–carbon bond formation between vinylchromium compounds and aldehydes was first described by Nozaki and co-workers (ref. 25). Extensive investigation involving vinylchromium reagents was subsequently carried out by Kishi and co-workers (ref. 26). This pivotal step based on the Nozaki–Kishi coupling reaction would generate the indolizidine framework, install the (*E*)-alkylidene side chain, and establish the C-7 hydroxy stereochemistry in a single operation.

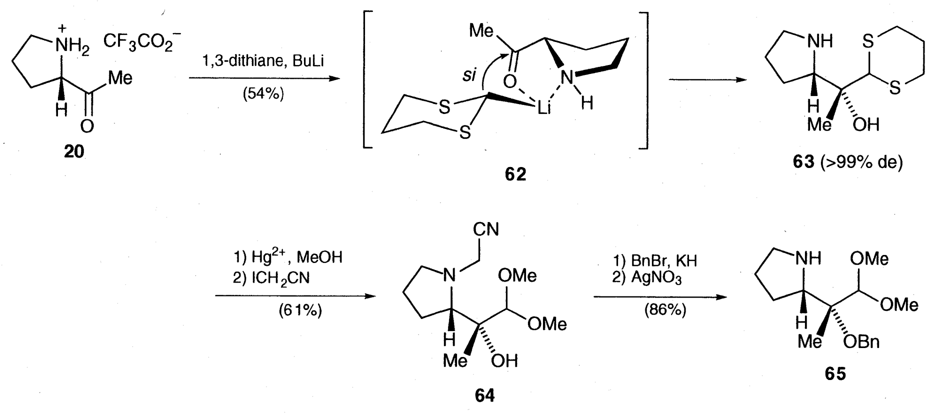


60 61

(3)

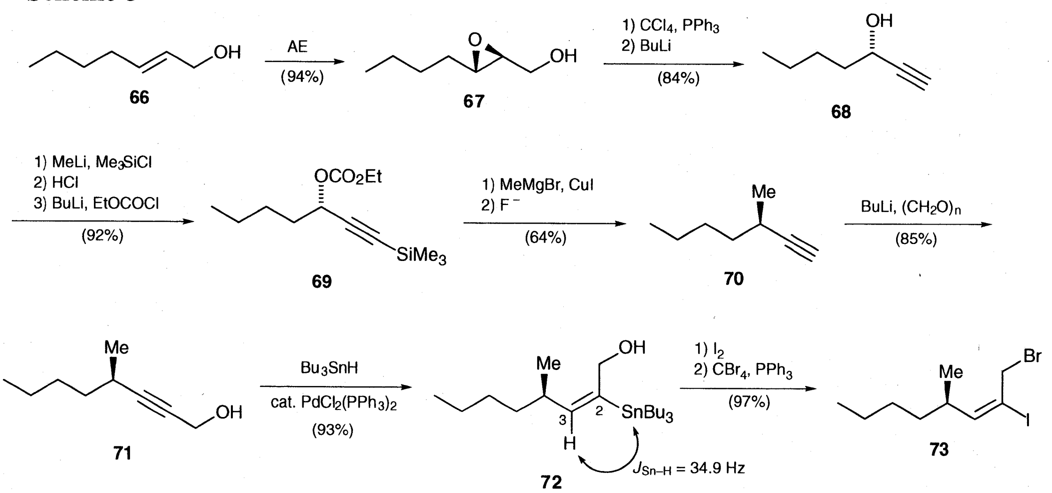
4.2 Total Synthesis of (+)-Allopumiliotoxin 267A. Wc initially targeted the enantioselective approach to the synthesis of (+)-allopumiliotoxin 267A (**10**) utilizing the intramolecular chromium-mediated cyclization based on the Nozaki–Kishi reaction. To this end, our efforts were directed toward the preparation of the cyclization substrate, which could be disconnected to give the pyrrolidine and side chain fragments. The required pyrrolidine unit **65** was prepared from the trifluoroacetate salt (**20**) of (*S*)-2-acetylpyrrolidine as outlined in Scheme 7. Thus, **20** was treated with 2-lithio-1,3-dithiane to produce the tertiary alcohol **63** as a single diastereomer with generation of the desired chirality according to Cram's cyclic model **62**. After acetal exchange ($Hg(ClO_4)_2$, MeOH) and N-protection by the cyanomethyl group, O-benzylation of the tertiary alcohol (BnBr, KH) followed by deblocking of the cyanomethyl group ($AgNO_3$) was carried out to form the pyrrolidine fragment **65**.

Scheme 7



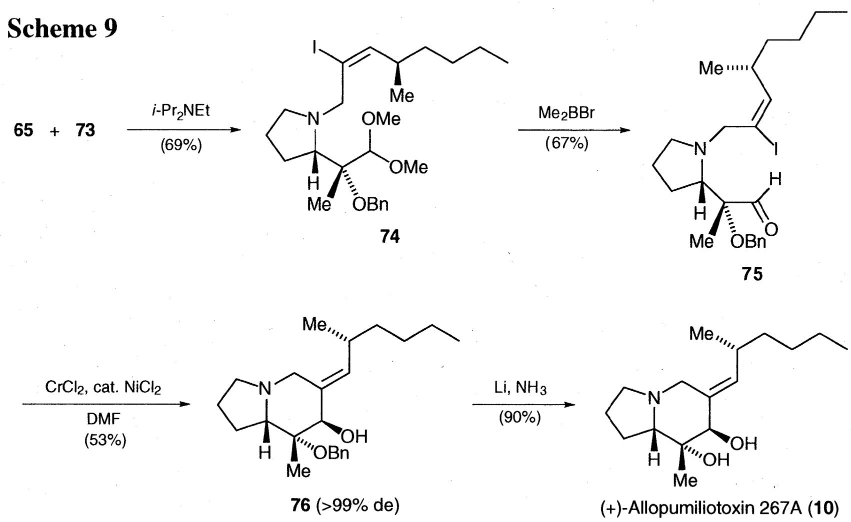
We next turned our attention to elaboration of the alkenyl iodide side chain fragment **73** as shown in Scheme 8. Asymmetric epoxidation of 2-hexenol (**66**) using diethyl L-tartrate gave the epoxide **67**, which was converted to (*S*)-1-heptyn-3-ol (**68**) under the conditions developed in Takano's laboratory (ref. 27). Following Overman's method (ref. 28), the heptynol **68** was converted to the (*R*)-alkyne **70**, which underwent hydroxymethylation with BuLi and paraformaldehyde to afford the propargyl alcohol **71**. Stereospecific construction of the *E* geometry was successfully achieved by applying palladium-catalyzed syn hydrostannylation to **71**, affording regioselectively the 2-stannyl adduct **72**. The anticipated *E* regiochemistry of **72** was verified by the small coupling constant (34.9 Hz) between Sn-2 and H-3. Subsequent iododestannylation with iodine followed by bromination (CBr_4 and PPh_3) yielded the vinyl iodide **73** with complete retention of the *E* geometry.

Scheme 8

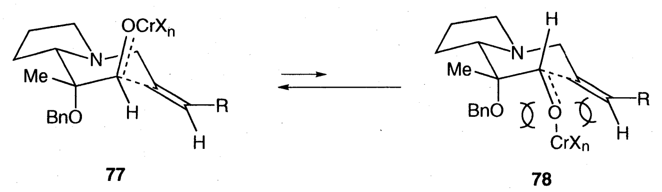


N-Allylation of the pyrrolidine **65** with the allyl bromide **73**, followed by acetal cleavage afforded (*E*)-iodoalkenyl aldehyde **75** (Scheme 9). Intramolecular Ni(II)/Cr(II)-mediated cyclization of **75** smoothly proceeded under mild conditions (DMF, room temperature), giving rise to **76** with the required axial C-7 hydroxy group with no evidence of the C-7 epimer in the ^{1}H NMR spectrum. Completion of the synthesis of (+)-allopumiliotoxin 267A (**10**) was accomplished via reductive cleavage of the benzyl group of **76** under the Birch conditions.

Scheme 9



The extremely high degree of diastereoselectivity in this process can be explained by a chair-like transition state (eq 4). Cyclization conformer **78** is destabilized by allylic 1,3-strain between the quasi equatorial chromium alkoxide and the alkene and, more importantly, by steric hindrance and electrostatic repulsion between the benzyloxy and the chromium alkoxide group bearing a partial negative charge. Neither of these interactions exists in the preferred transition structure **77**.

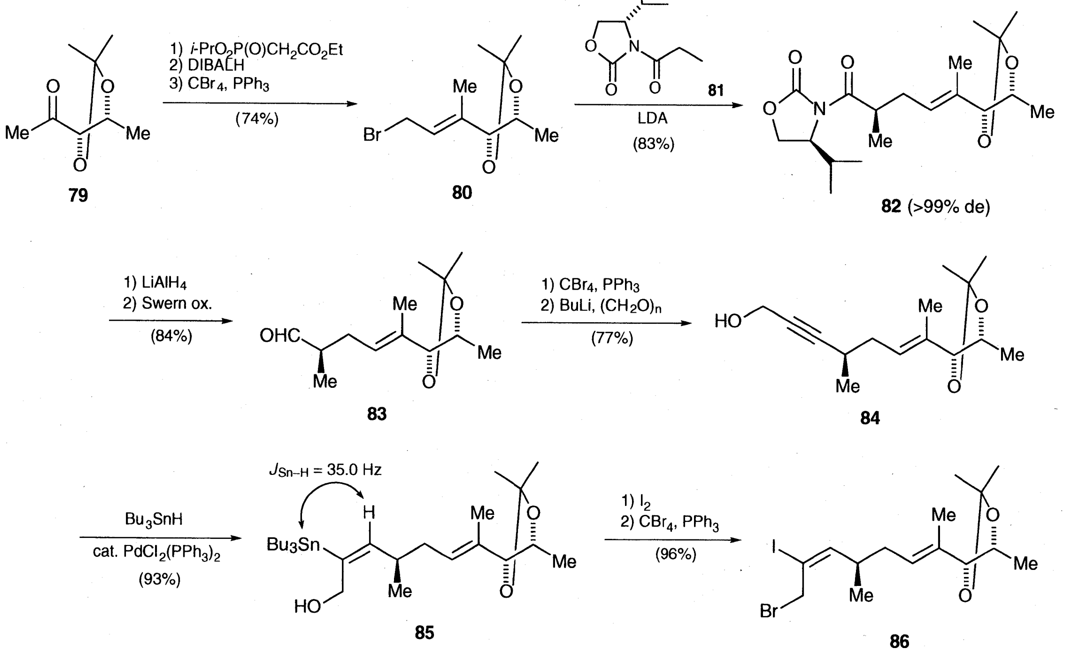


(4)

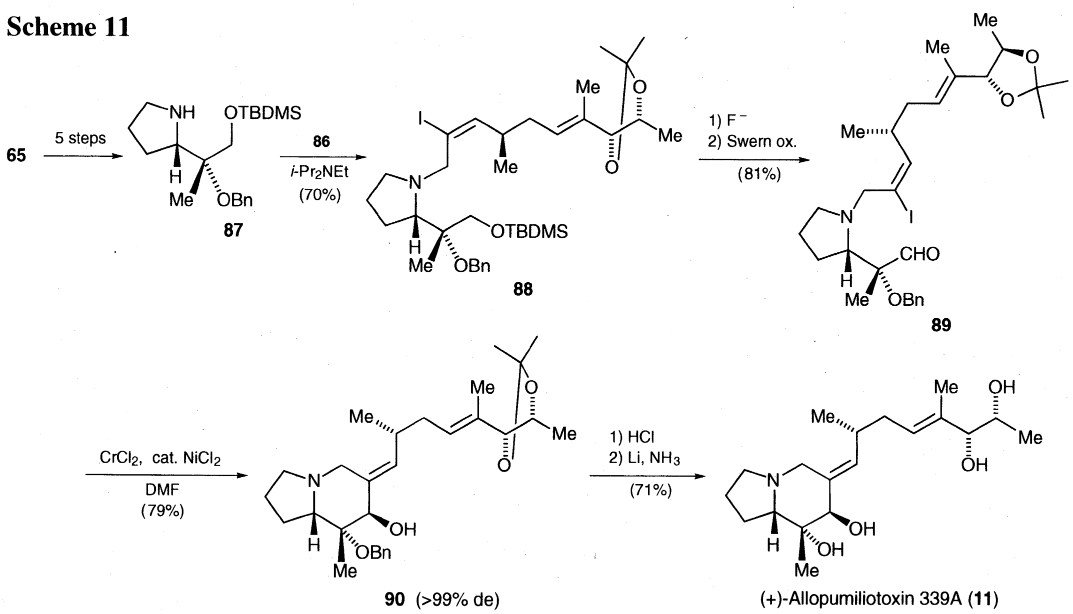
4.3 Total Synthesis of (+)-Allopumiliotoxin 339A. We further investigated extension of the above strategy to the preparation of allopumiliotoxin 339A (**11**). Toward this end, the sequence began with the elaboration of the alkenyl iodide side chain fraction **86** as depicted in Scheme 10. Thus, the methyl ketone **79** was transformed into the allyl bromide **80** in a straightforward manner involving Horner–Emmons condensation. Evans alkylation of the (*S*)-oxazolidinone derivative **81** with **80** provided **82** with virtually complete diastereoselection. After reductive removal of the oxazolidinone auxiliary on **82** ($LiAlH_4$) followed by Swern oxidation, the resulting aldehyde **83** was converted to the propargyl alcohol **84**. In a manner similar to that described above for the preparation of **73**, **84** was converted to the (*E*)-vinyl iodide **86** in three steps in stereospecific and highly regioselective manners.

N-Allylation of the pyrrolidine **87**, available in 5 steps from the pyrrolidine fraction **65** used for the synthesis of (+)-allopumiliotoxin 267A (**10**), with the allyl bromide **86** afforded **88**, which was converted to the (*E*)-iodoalkenyl aldehyde **89** (Scheme 11). On treatment of **89** with Ni(II) and Cr(II), intramolecular coupling proceeded with virtually complete stereoselection to give **90**, whereby the indolizidine framework with the (*E*)-alkylidene side chain and the C-7 hydroxy group with the correct stereochemistry could be assembled in a single operation. The same stereochemical argument as described for the chair-like transition state **77** should hold for this process. Cleavage of the isopropylidene group and subsequent reductive cleavage of the benzyl group provided (+)-allopumiliotoxin 339A (**11**) in 71% yield.

Scheme 10



Scheme 11



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